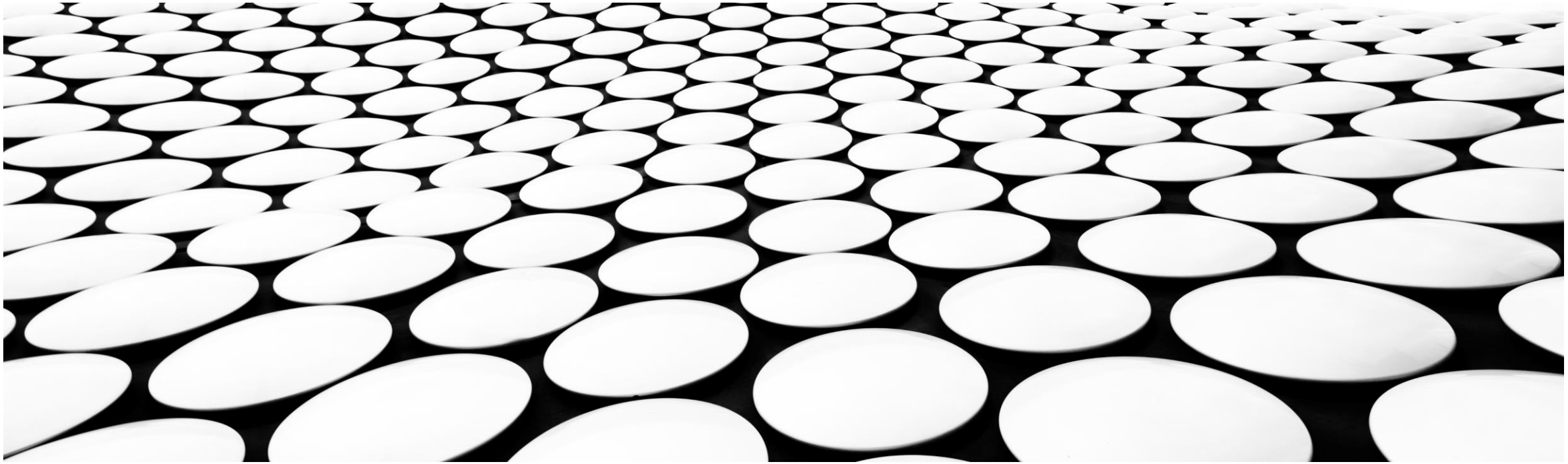


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Endpoint Security

Engineering Cybersecurity Solutions

Donnacha Forde



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$whoami

- Developer
- Architect
- Manager
- Technical Director
- Technology Strategist



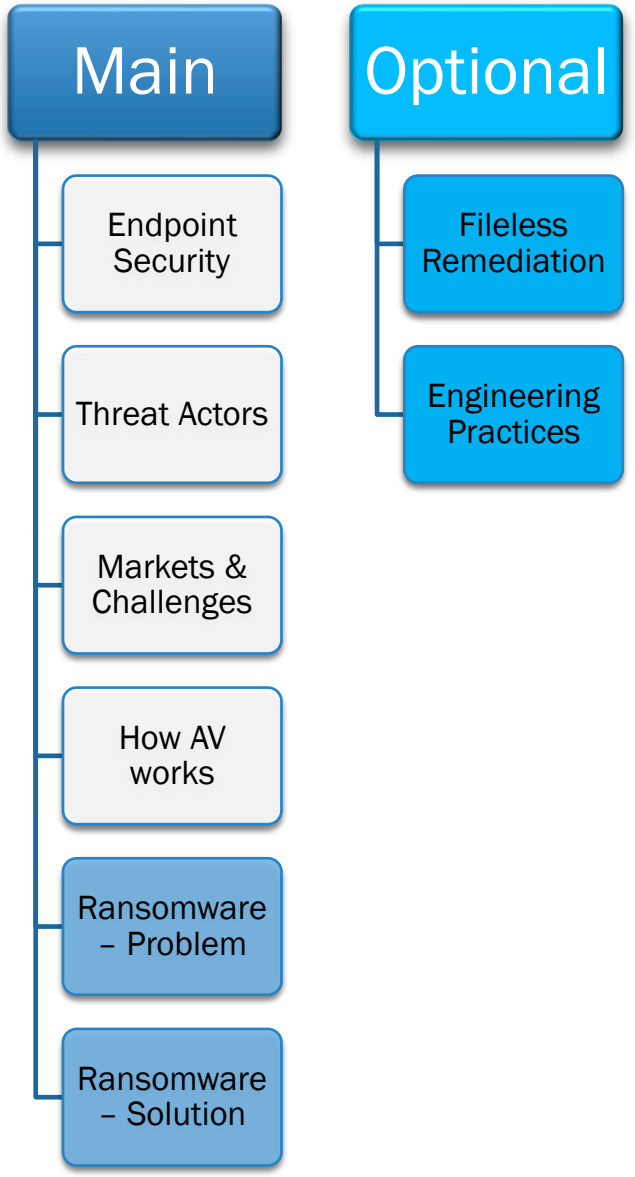
Mostly Finance (~10yrs)

Mostly Systems-Management (~15yrs)

Security (~10 yrs)

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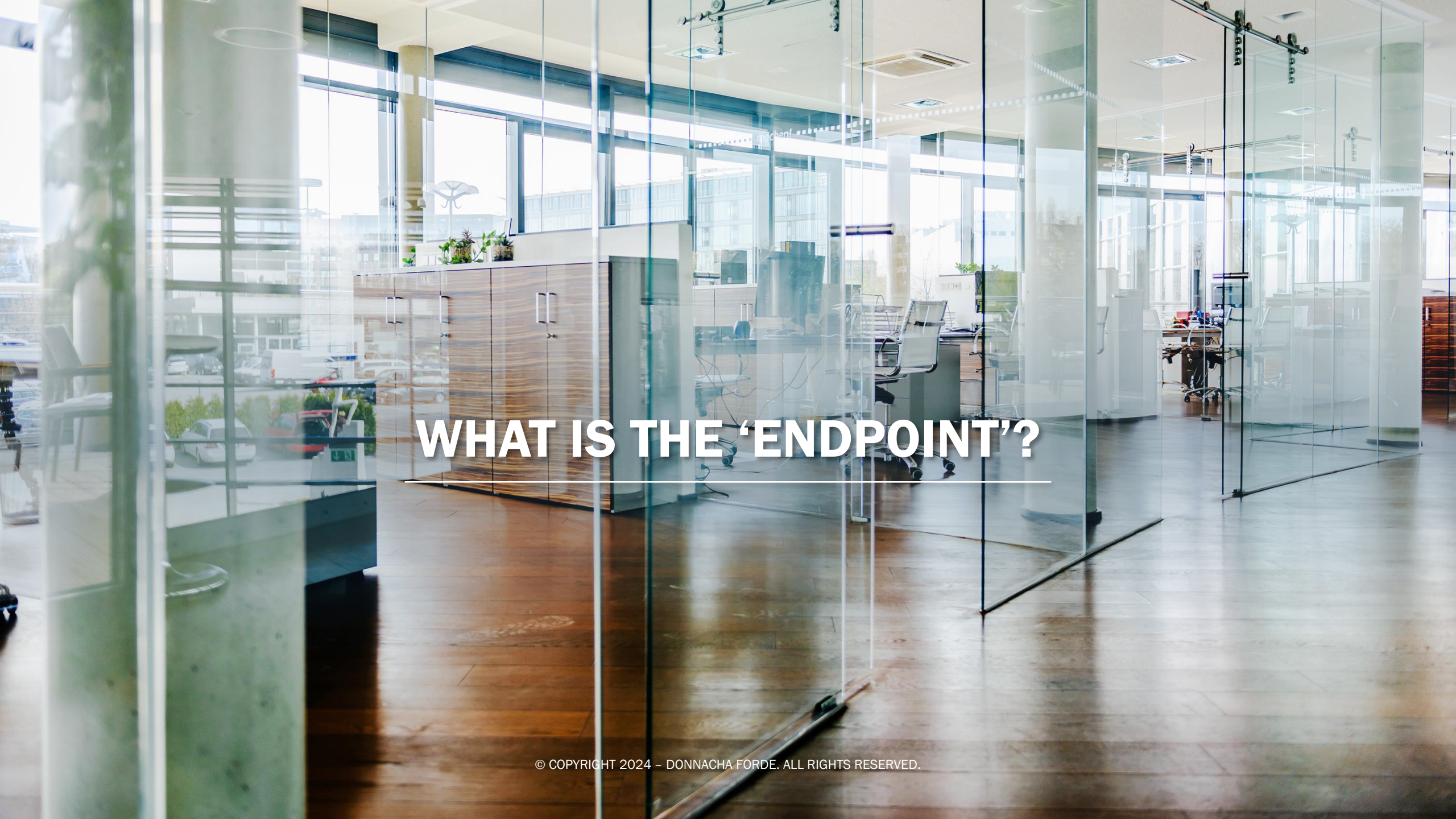
AGENDA



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Endpoint Security

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WHAT IS THE ‘ENDPOINT’?

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THE ENDPOINT



Desktop PC



Laptop



Tablet



Mobile



Server



Gaming Device



IoT Device



Smart Home Devices



VMs & Containers



Network

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ECOSYSTEMS



Windows



macOS



Linux



ChromeOS



iOS



Android



x64



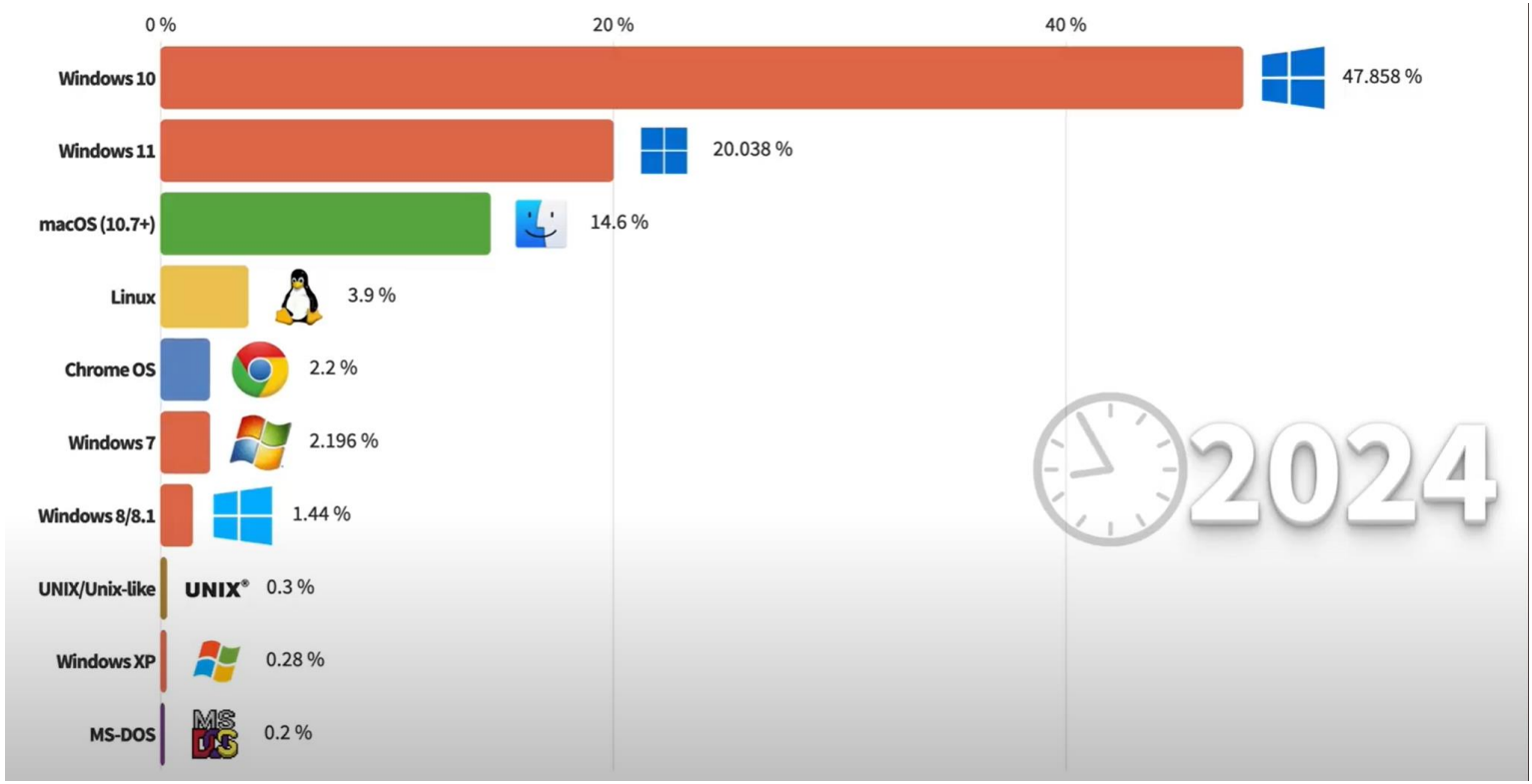
arm64

Reference: https://donnachaforde.github.io/blogs+posts/software-engineering/wouldnt-start-here/

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OS Prevalence – Market Share

- Windows 10 & 11 >67%
- macOS ~15%
- Linux ~4%
- ChromeOS ~2%



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Source: https://www.youtube.com/watch?v=Tjc-qrjzRcU

WHAT IS ENDPOINT SECURITY?



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What is Endpoint Security?

Definition: Securing network-connected devices from cyber-threats.



Endpoints are prime targets for cyberattacks

Ransomware, Phishing, Identity



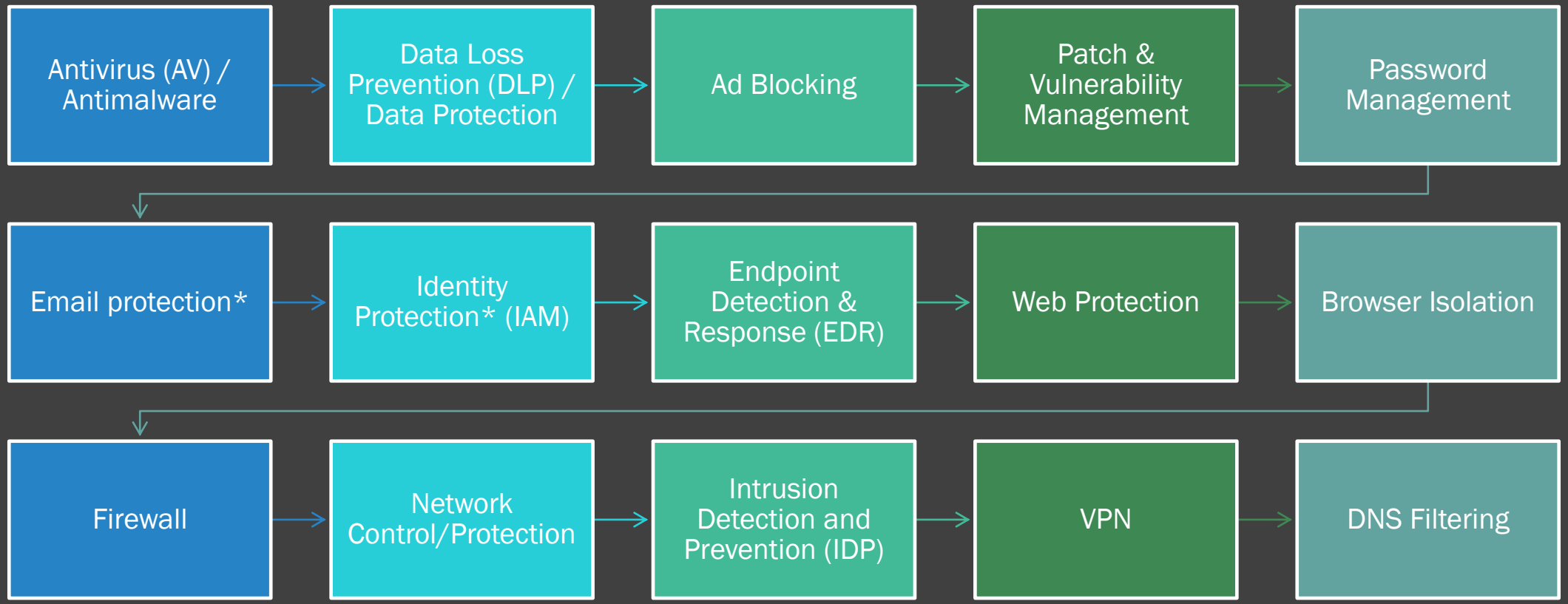
Modern Day Factors

Shift to cloud computing

Increased working from home

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COMPONENTS OF ENDPOINT SECURITY



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*also involve Server/Cloud Technology

Core Endpoint Technologies

EPP – Antivirus and Antimalware

- Detection
  - Signature-based Detection → Heuristics → Rule-based → Behaviour-based
- On demand scanning/On access scanning
- Real-time Protection
  - Monitor Files/Processes, Downloads and Network Traffic
- Quarantine
- Remediation
- Anti-ransomware
- Internal
  - Self-Protection/Tamper Detection
  - Trust → Performance

Endpoint Detection & Response

- Real-time Monitoring
- Visibility
- Detection
  - Automatic via Content Rules
  - Manual by SOC Analysts
- Analysis
  - Historic Queries
  - Distributed Queries
- Response
  - Remote Access
  - Network Isolation
- Continuous Data Collection → SOC Analysts/Threat Hunting

Reference: https://donnachaforde.github.io/blogs+posts/software-engineering/endpoint-protection/

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EPP versus EDR

Natural Tension:

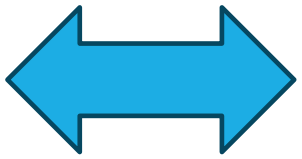
Protection v Detection



EPP

Prevention

Blocking





EDR

Monitoring

Detection

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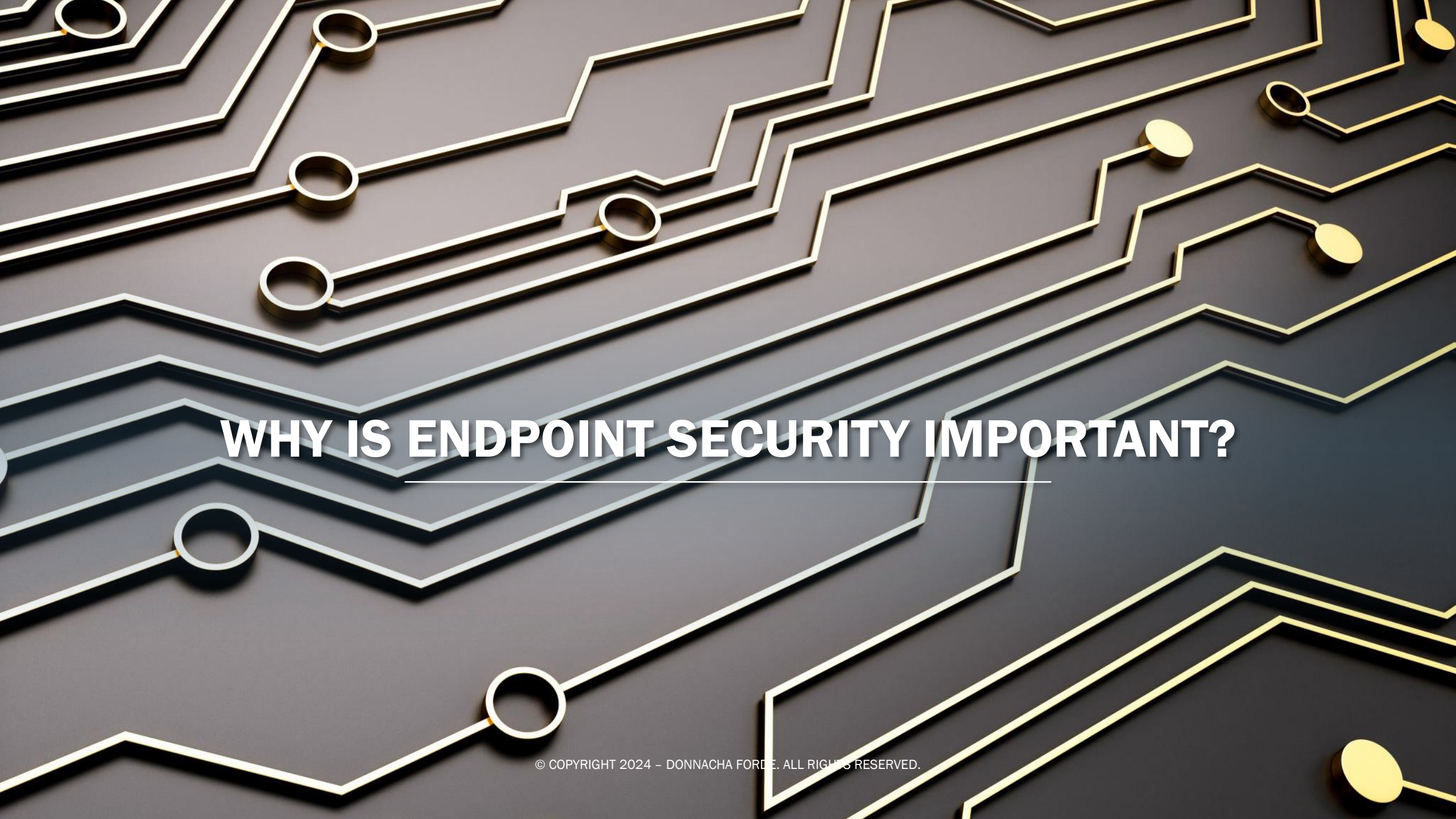
Cloud Infrastructure

- Systems Management
  - Multi-tenant Cloud
  - Managed Solutions (e.g. MDR)
  - Large no. of users
  - Large no. of endpoints
  - Huge volume of events

- Endpoint Support System
  - Client/Server solutions
  - Cloud-based Solutions
    - e.g. ML Models, Behavioural Analysis, Detonation/Sandbox
  - Event Management
    - Messaging
    - Queue Management
    - Persistence (e.g. Graphs)
  - IPC
    - Distributed Queries (Cloud → Endpoint)
    - Lookup (Endpoint → Cloud)
  - Performance ←→ POP



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WHY IS ENDPOINT SECURITY IMPORTANT?

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WHY IS ENDPOINT SECURITY STILL SO CRITICAL?



Ever increasing number of connected devices



Industry trends

Bring your own device (BYOD)

Shadow IT

Remote Working



Prevalence of Real-world threats

Ransomware

Data breaches

Insider-threats



Cost of Security Incidents

Direct extorsion - ransom

Financial loss – impact on business function

Reputational damage

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Costs of Cybercrime

- Estimated cost globally
  - 2021 → > $6 Trillion
  - 2022 → > $8 Trillion
  - 2023 → > $11 Trillion
  - 2026 → $20 Trillion

- $6 Trillion in 2021 → World's third-largest economy (after US and China)
  - Up from $3 Trillion in 2015
- Context
  - Greatest transfer of economic wealth in history
  - Greater costs than damage from natural disasters
  - More profitable than Global Trade in Illegal Drugs

Source: https://www.evolvesecurity.com/blog-posts/actual-cost-of-cybercrime

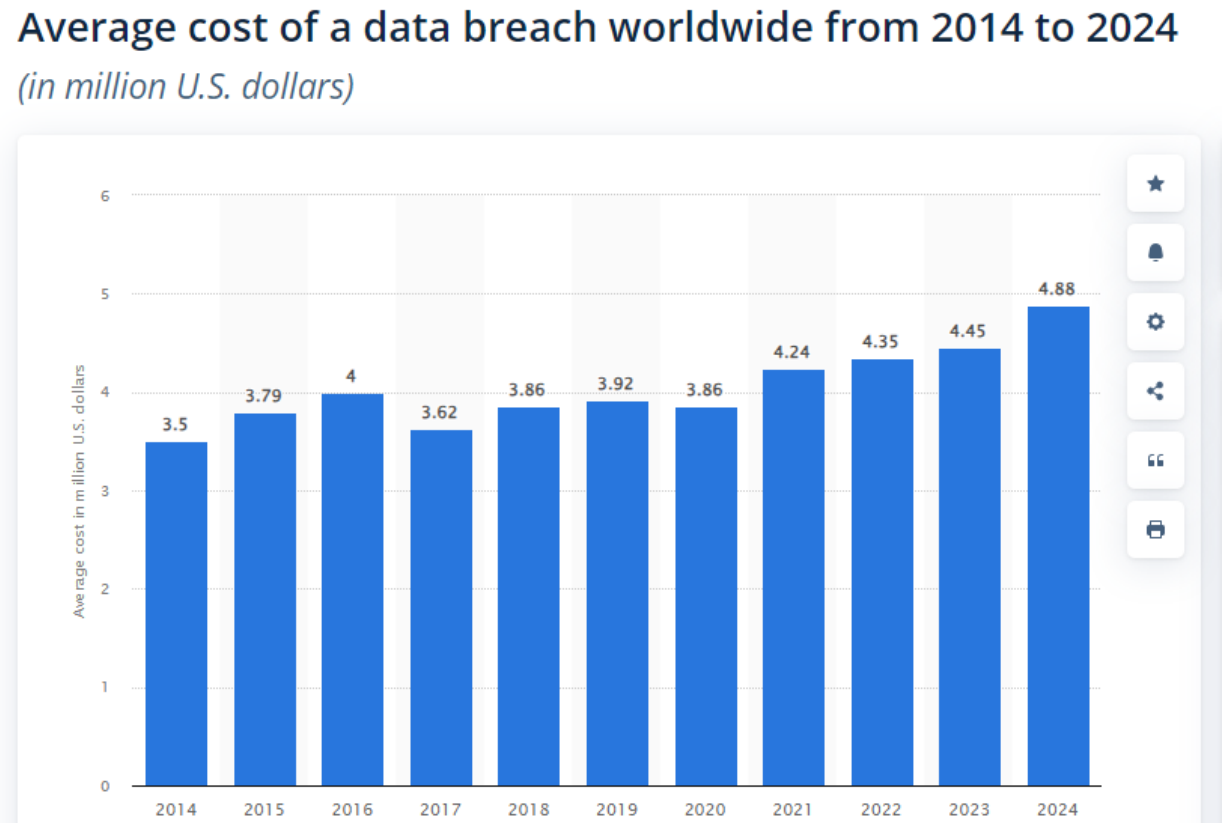
Source: https://cybersecurityventures.com/hackerpocalypse-cybercrime-report-2016

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Costs of Data Breaches

- Estimated 70% of successful data breaches originate at the endpoint
- Data Breach Costs
  - Average cost of data breach – all sectors: $4.88M (10% increase over 2023)
  - Average cost of data breach in the financial sector: $6.08M (3% increase over 2023)

Source: https://www.evolvesecurity.com/blog-posts/actual-cost-of-cybercrime



Source: https://www.statista.com/statistics/987474/global-average-cost-data-breach

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Adversaries

Who are the threat actors?

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Where are the threats coming from?

Top three threats



Organized Crime

Criminal gangs

Often state-sponsored and/or state sanctioned



Nation-state

Political and/or Financial motives



Hacktivism

Disruption

Exposure

Access Data

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Famous Chollima



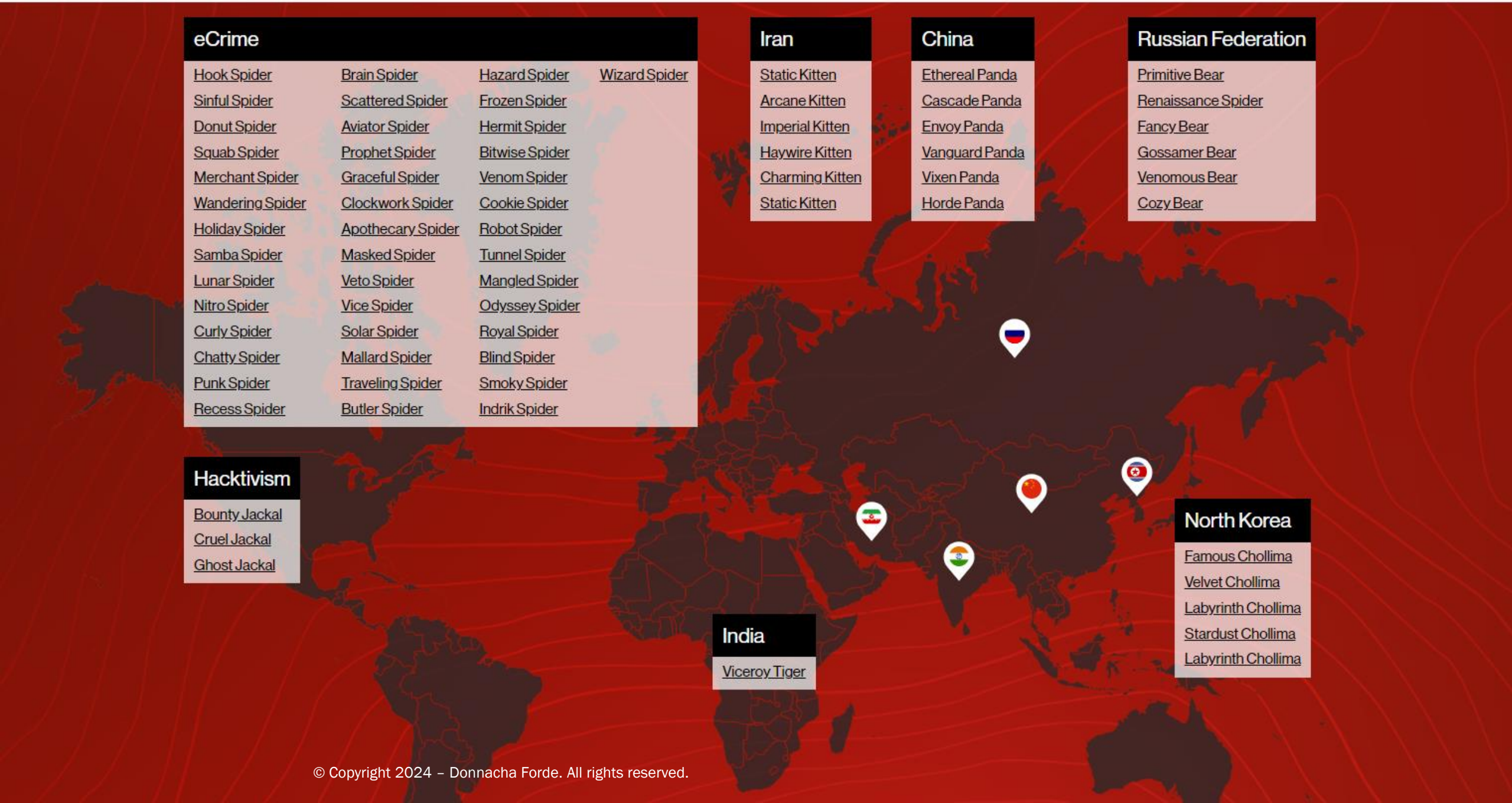
Punk Spider



Horde Panda

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Global Threat Landscape



eCrime

- Hook Spider
- Sinful Spider
- Donut Spider
- Squab Spider
- Merchant Spider
- Wandering Spider
- Holiday Spider
- Samba Spider
- Lunar Spider
- Nitro Spider
- Curly Spider
- Chatty Spider
- Punk Spider
- Recess Spider
- Brain Spider
- Scattered Spider
- Aviator Spider
- Prophet Spider
- Graceful Spider
- Clockwork Spider
- Apothecary Spider
- Masked Spider
- Veto Spider
- Vice Spider
- Solar Spider
- Mallard Spider
- Traveling Spider
- Butler Spider
- Hazard Spider
- Frozen Spider
- Hermit Spider
- Bitwise Spider
- Venom Spider
- Cookie Spider
- Robot Spider
- Tunnel Spider
- Mangled Spider
- Odyssey Spider
- Royal Spider
- Blind Spider
- Smoky Spider
- Indrik Spider
- Wizard Spider

Iran

- Static Kitten
- Arcane Kitten
- Imperial Kitten
- Haywire Kitten
- Charming Kitten
- Static Kitten

China

- Ethereal Panda
- Cascade Panda
- Envoy Panda
- Vanguard Panda
- Vixen Panda
- Horde Panda

Russian Federation

- Primitive Bear
- Renaissance Spider
- Fancy Bear
- Gossamer Bear
- Venomous Bear
- Cozy Bear

Hacktivism

- Bounty Jackal
- Cruel Jackal
- Ghost Jackal

India

- Viceroy Tiger

North Korea

- Famous Chollima
- Velvet Chollima
- Labyrinth Chollima
- Stardust Chollima
- Labyrinth Chollima

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Market & Challenges

Who are we protecting and what are the challenges?

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Market Segments



Enterprise

Primary Target for Adversaries

Environment

- Exist behind Firewall
- Network traffic goes through a gateway

Managed Endpoint

- More sterile environment, Less deviation



SME

Enterprise v SOHO



Consumer

More diverse environment

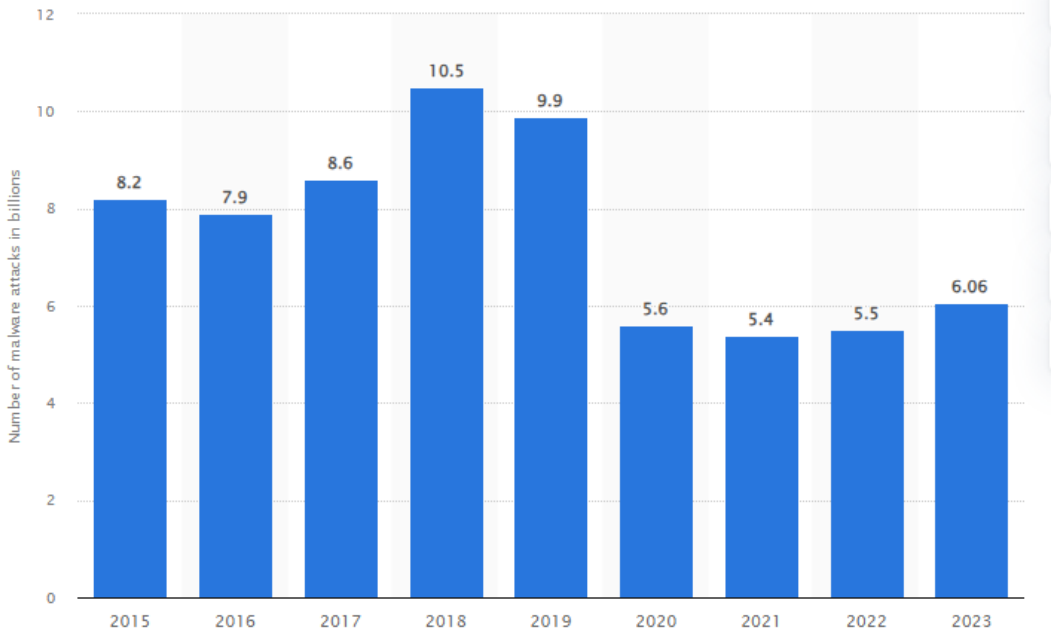
Not sterile

Greater exposure

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Annual Number of Malware Attacks

- Period: 2015 – 2023
- 6.06 billion in 2023
- 10% increase on previous year
- 10.5 billion attacks in 2018
- Websites are the most common attack vector



Source: https://www.statista.com/statistics/873097/malware-attacks-per-year-worldwide/

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Endpoint Security Challenges

- Evolving Threats
  - Zero-day Attacks
  - Advanced Persistent Threats (APT)
  - Insider Threats
- Trends
  - Shadow IT
  - Remote Working
  - BYOD
- Complexity – Managing at Scale
  - Security & IT Management
- Reaction Time vs Volume of Data
  - SOC Analyst Response
  - Threat Hunting
  - Threat Intel



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Endpoint Engineering Challenges

- Scale
  - Volume of System Event Info
- Costs → COGS
  - CPU, Network, Storage
- Security vs User Convenience vs Performance
- Complex, large-scale environments
- Rapid Response
- Prompt Delivery of new features & solutions in a complex environment
- Risk & Impact



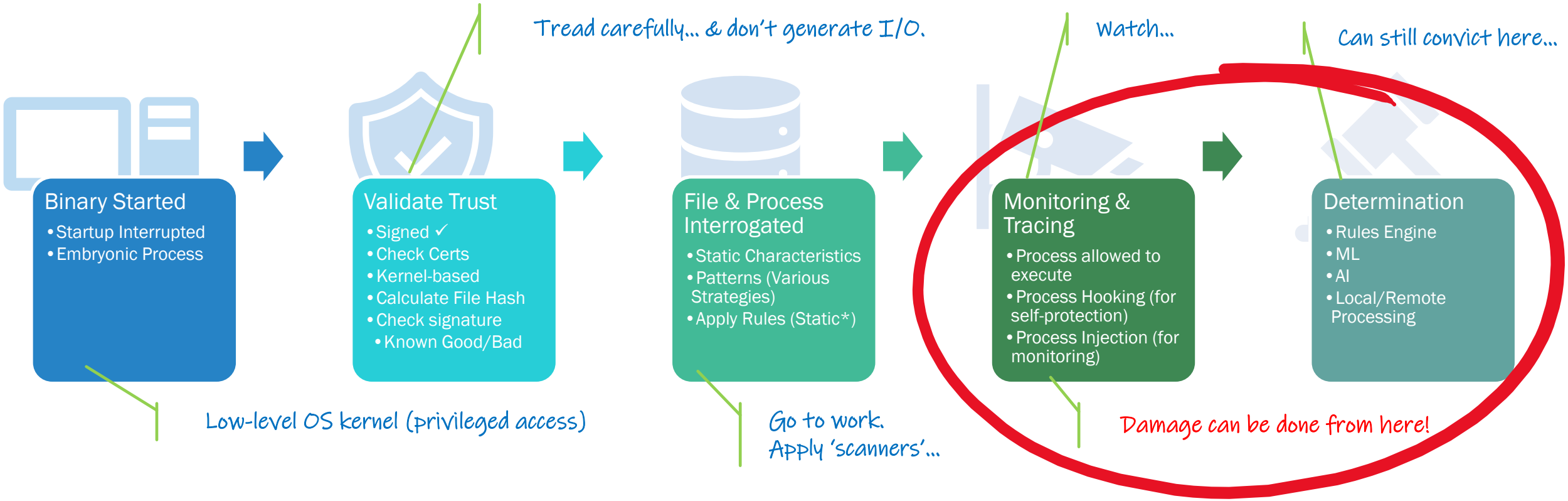
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Antivirus and Antimalware

How AV & Ransomware works

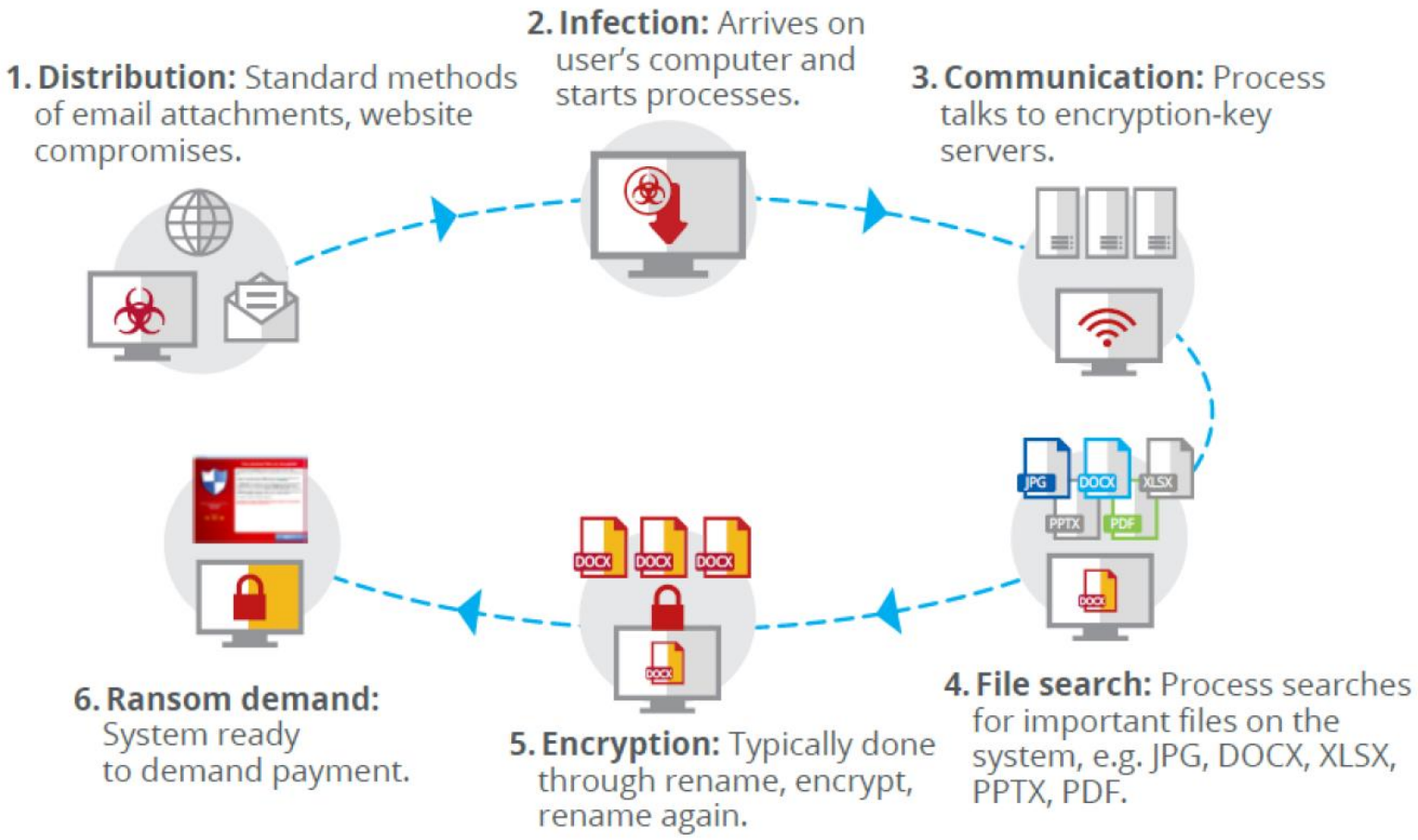
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How AV works



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How Ransomware Works



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Image Credit: McAfee Whitepaper – *You, Machine Learning & Zero-Day Malware*

McAfee Behavioural Analysis



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Image Credit: McAfee Whitepaper – *You, Machine Learning & Zero-Day Malware*

The Problem – Ransomware

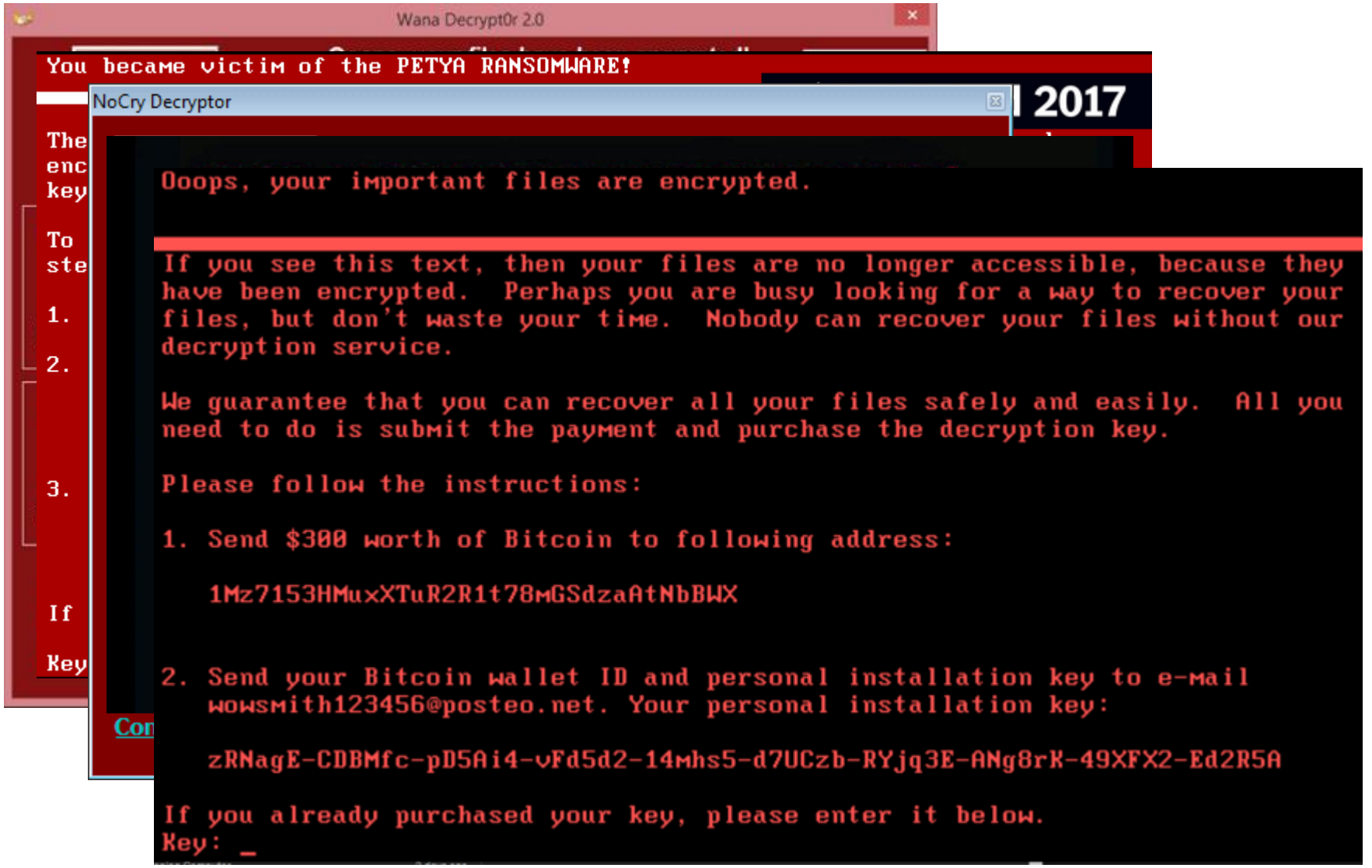
A look at Ransomware Trends

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Circa 2016

- Top 10 Worst Ransomware Attacks

1. NotPetya
2. WannaCry
3. Locky
4. CrySis
5. Nemucod
6. Jaff
7. Spora
8. Cerber
9. Cryptomix
10. Jigsaw



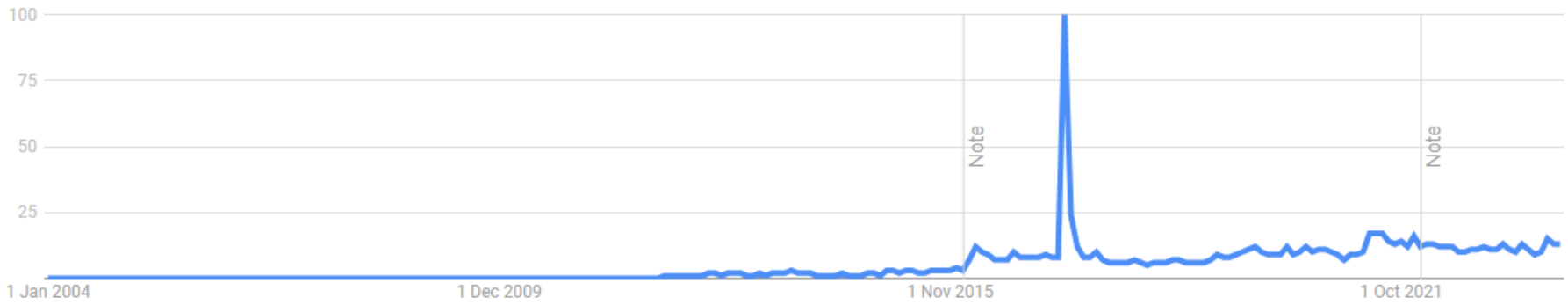
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ransomware
Search term

+ Compare

Worldwide | 01/01/2004 - 06/11/2023 | All categories | Web Search

Interest over time



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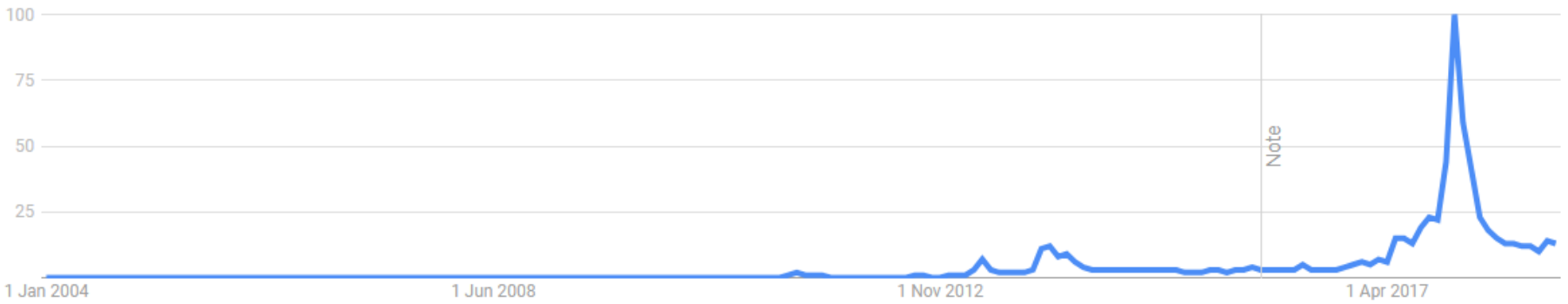
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Bitcoin
Search term

+ Compare

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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ransomware
Search term

Bitcoin
Search term

+ Add comparison

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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Ransomware continues to be a Threat

- US is the most attacked country.

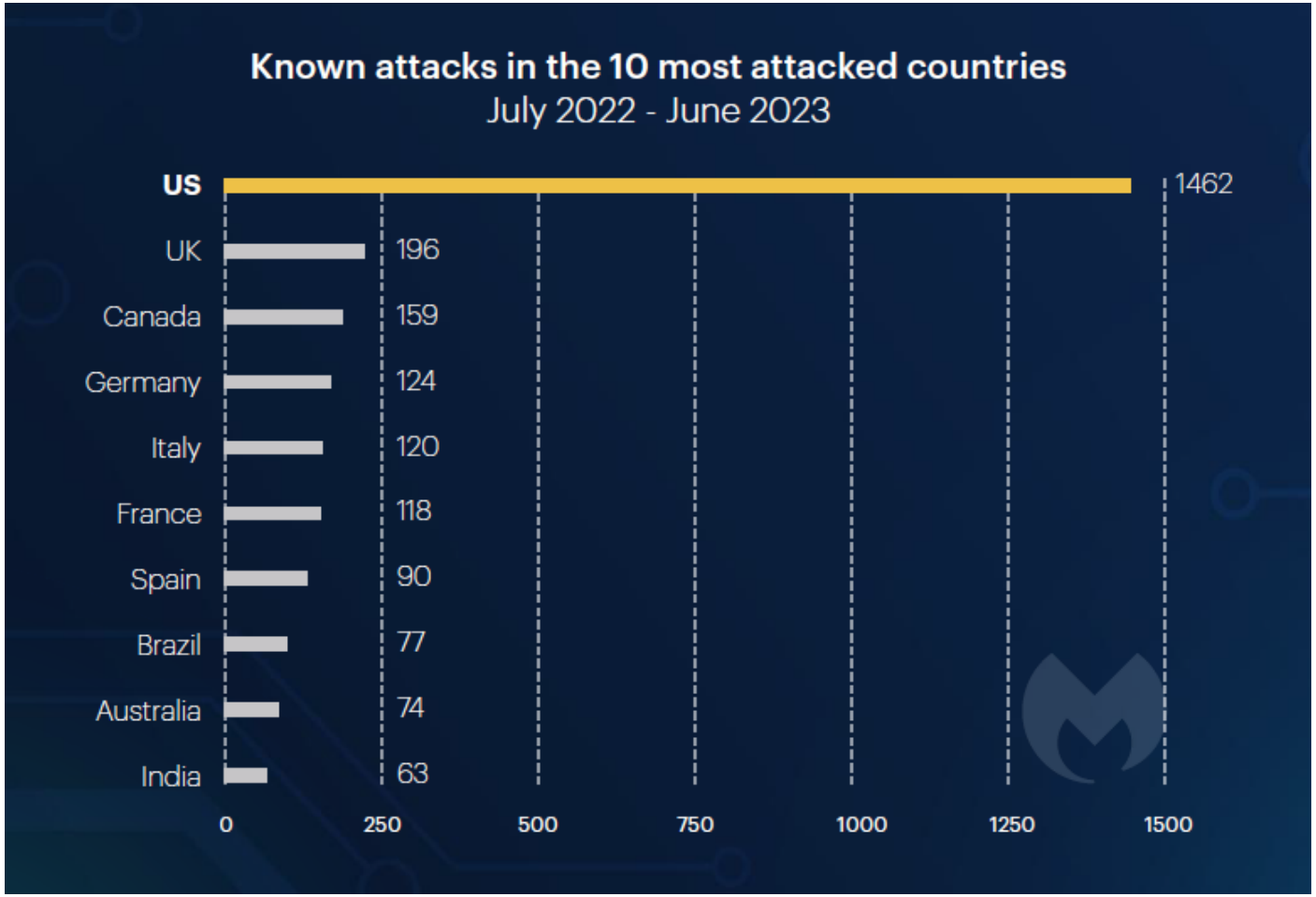


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Ransomware attacks in last 12 months.

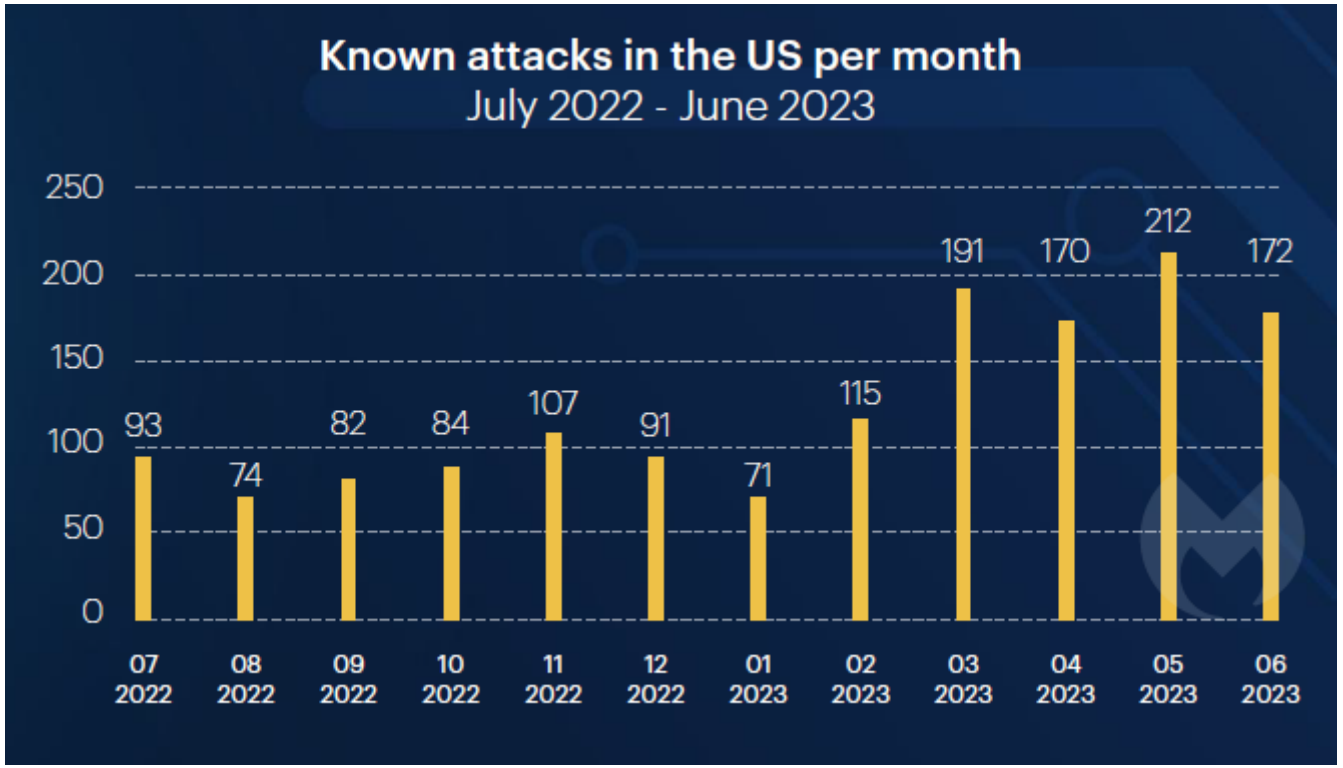


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Active gangs

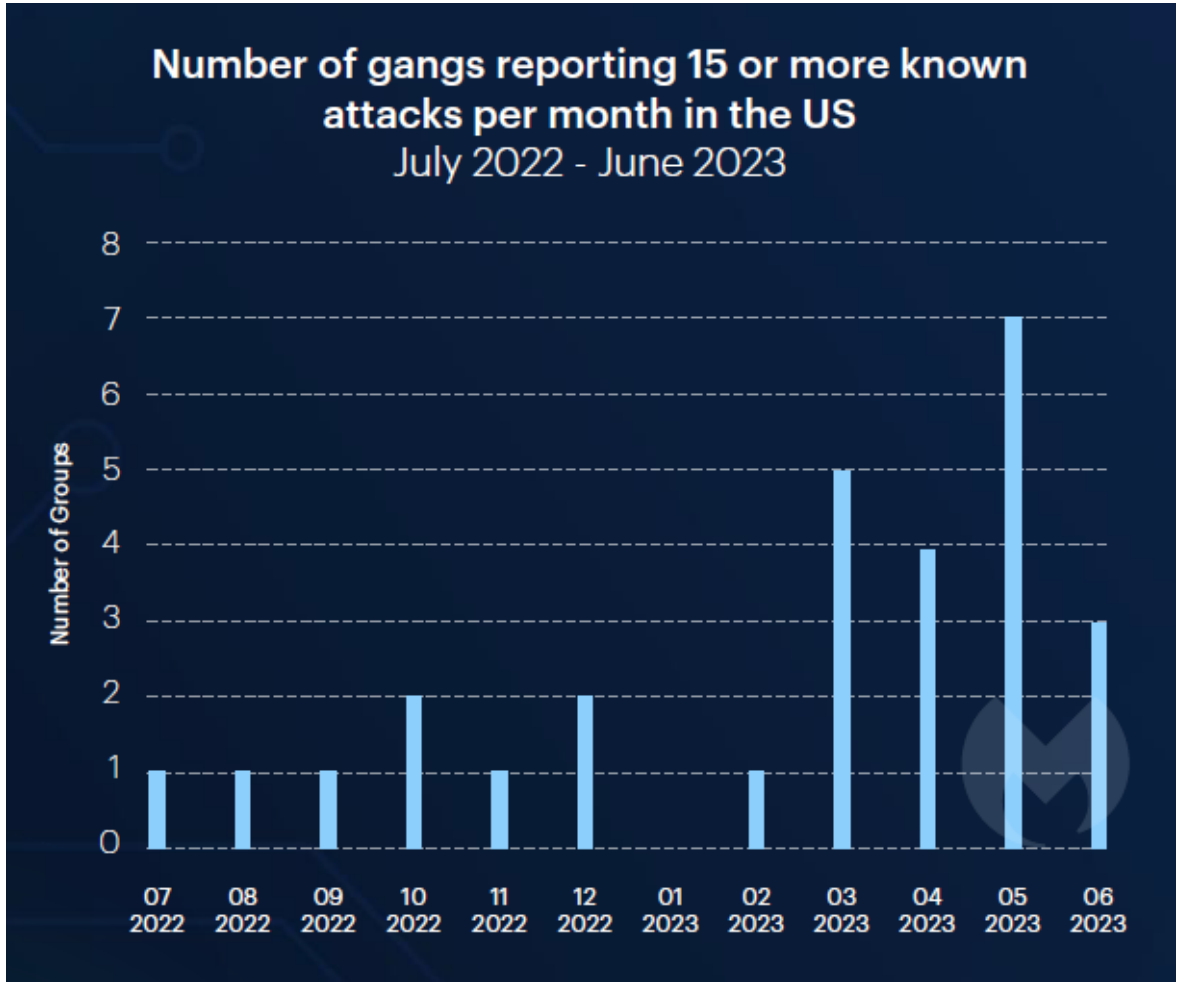


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Ransomware Gangs

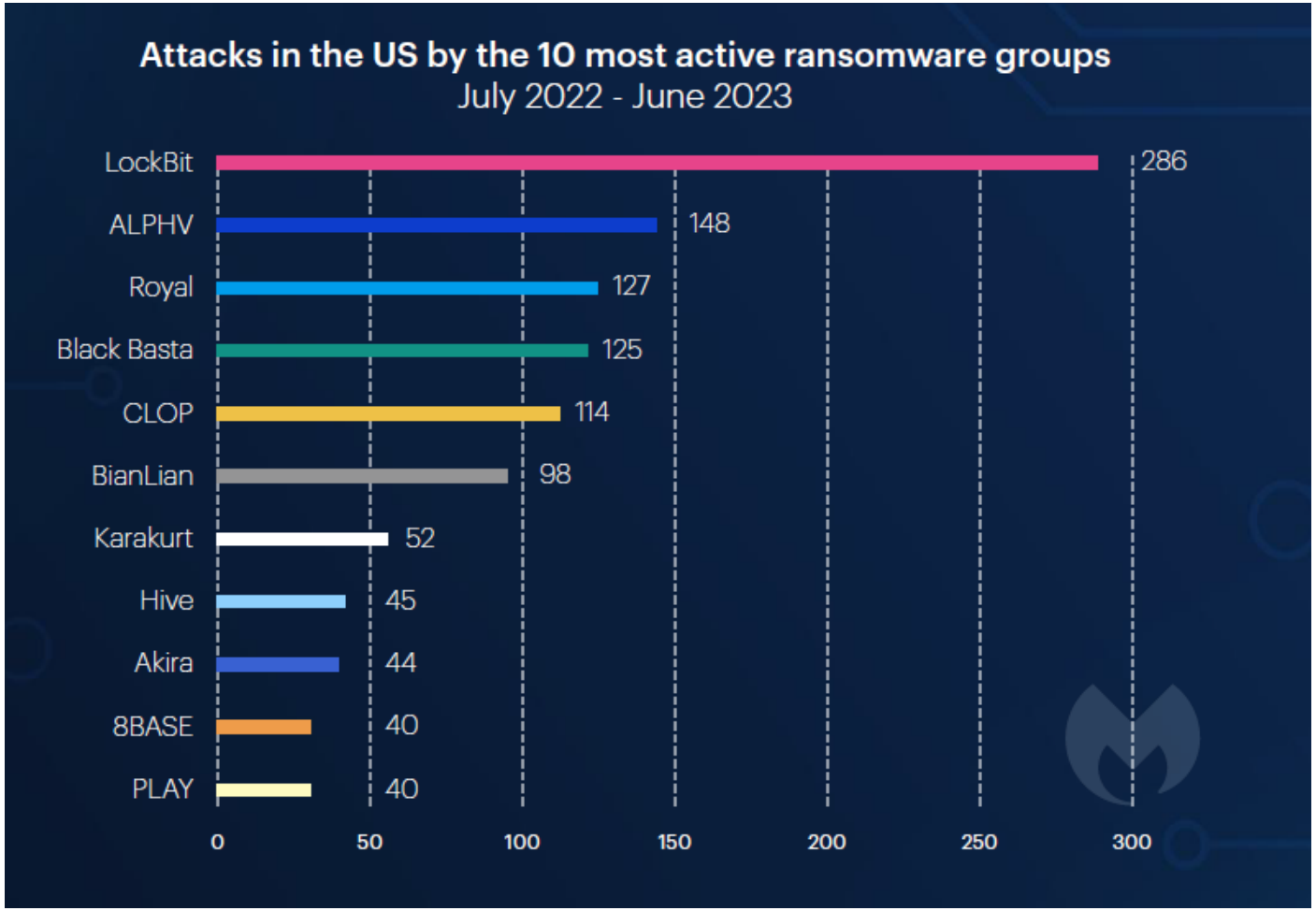


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Innovation

Devising an Anti-ransomware Solution

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Strategies



Prevention

Block Malware

Zero-day Ransomware?

Polymorphic Malware?



Avoidance

Cloud Storage

(e.g. Microsoft OneDrive)

Minimize impact → Restore to last known-good backup

Incur re-image cost & effort

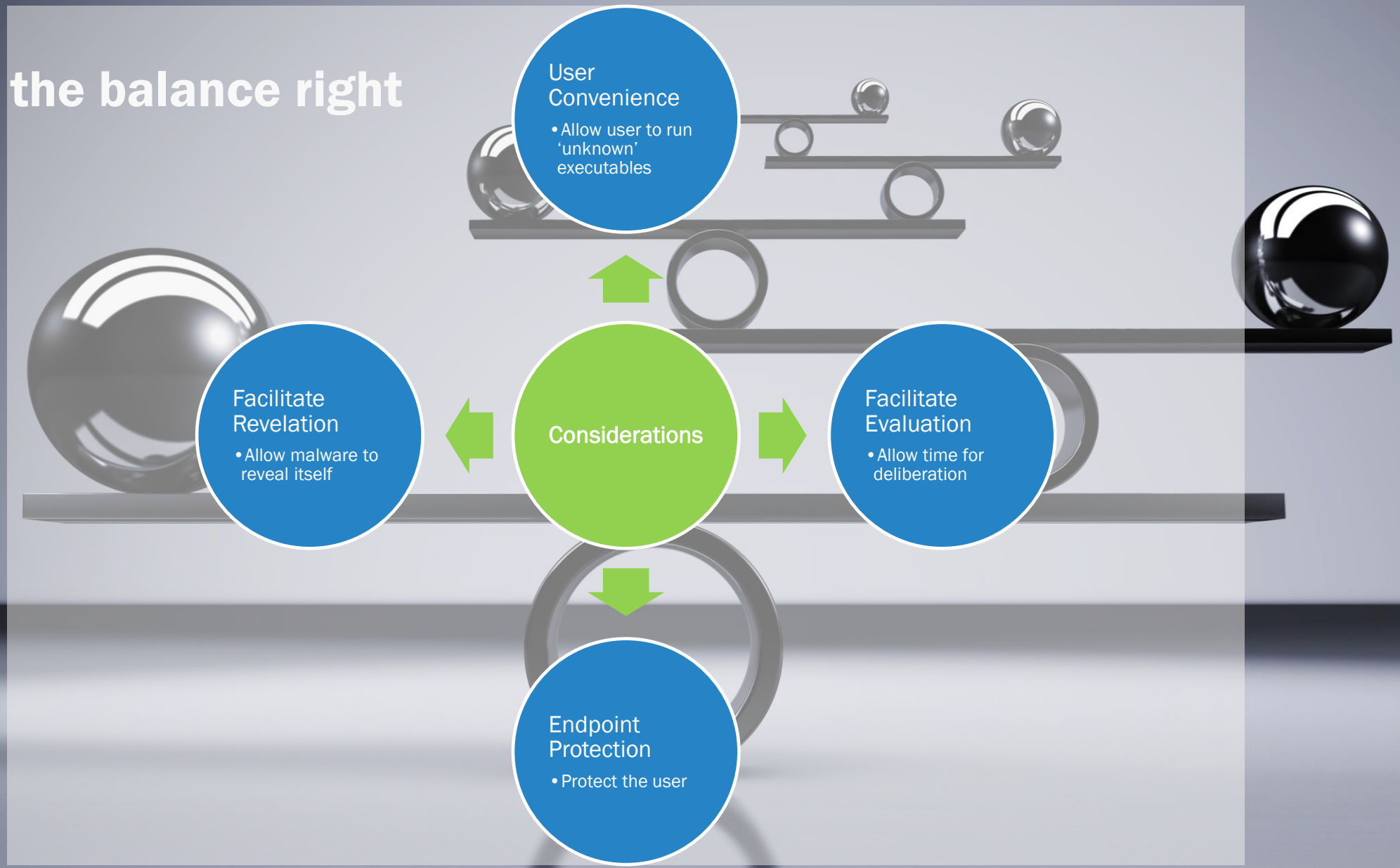


Remediate

Restore Endpoint to previous known-good state

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Getting the balance right



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REMEDIATION STRATEGIES



Confinement

Limit damage to a Virtual Image

Ditch VM on conviction

Roll forward if not convicted



Rollback Reversal

Perfect Undo

Rollback on conviction

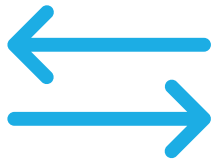
Do nothing if not convicted

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Remediation Strategies



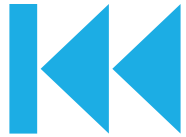
Optimistic v Pessimistic View



Roll forwards – or – Roll backwards



Most software != malware



Fewer instances where we need to take action → Roll backwards

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Rollback Remediation

What is it? How does it work?

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Rollback Remediation



Before Conviction

Monitor

Create Backups



Post Conviction

Kill Process Tree

Restore

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Rollback Remediation



Correction Technology

Does not detect

Does not protect



Passively Monitors

Performs Low-level Monitoring

Maintains ‘Process Genealogy’ Tree



Active Monitoring Triggered

Records detailed activity

Maintains copies of File, Directory and Registry changes.

Monitors child processes, proxies or any other surrogates.



Remediates

Terminates Process Tree

Uses Logs to restore all File, Directory & Registry changes & perform rollback.

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Remediation Details



File & Directory

Files & Directories restored to original state.

Encrypted files removed.

‘Dropped files’ removed.



Windows Registry

RegKeys & Values restored to original state.

Created RegKeys & Values removed.



File Attributes

Preserve Hidden/Archive file attributes.



Security Descriptors

Preserve file security metadata.

Restore File/Directory ownership.



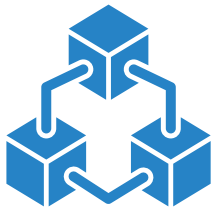
RunKeys & Startup Directories

Clean autostart registrations.

Clean known (and not-so-well known) directory & registry locations.

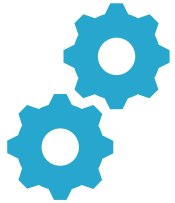
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Detecting Lateral Movement*



Windows Service Control Manager

Registration of background Windows Services (Daemons).



Windows Task Scheduler

Remove tasks scheduled for future execution.



Windows BITS

Background Intelligent Transfer Service

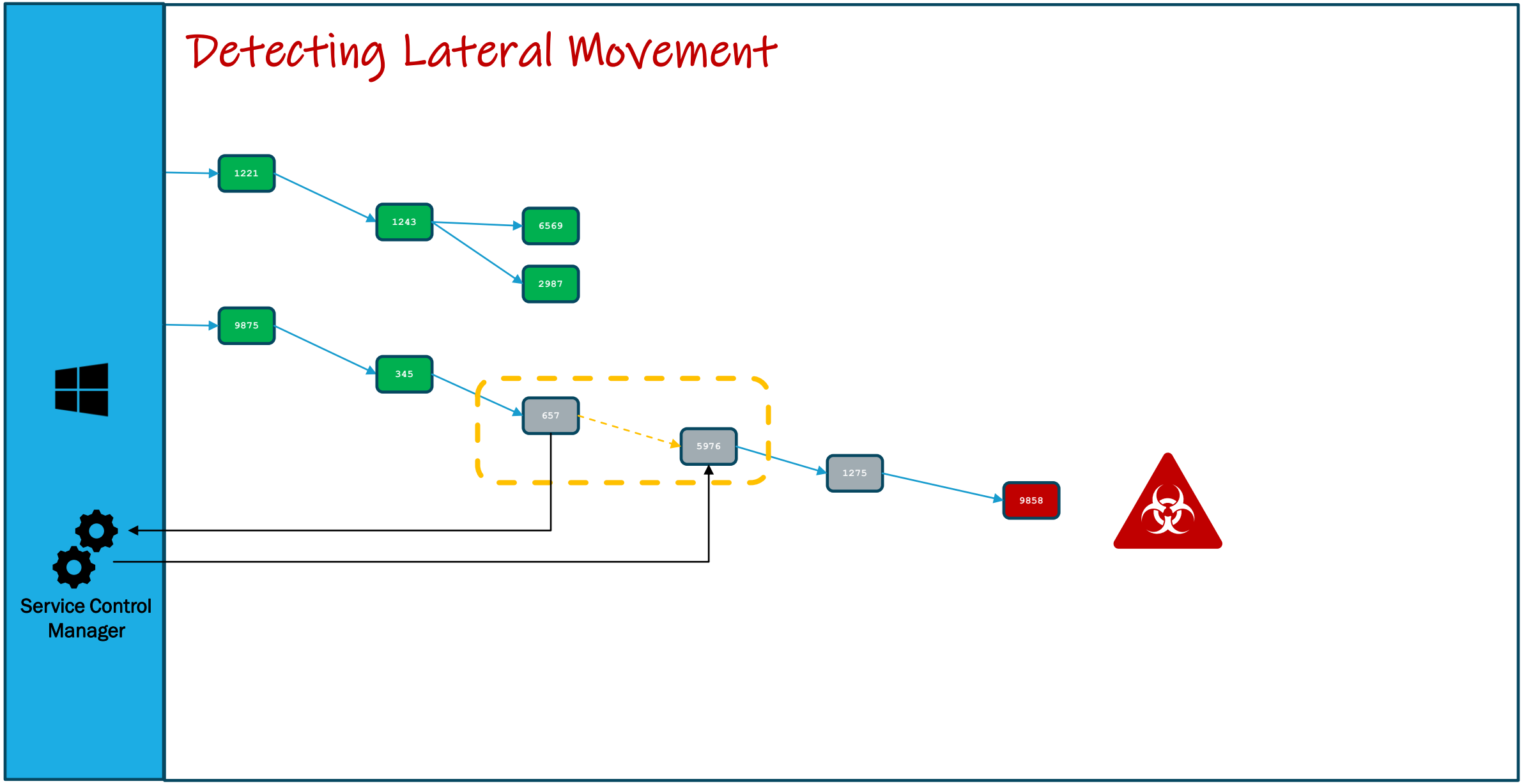
Facilitate intelligent file download

Download failure triggers registered handler.

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*Endpoint Lateral Movement (or Micro Lateral Movement

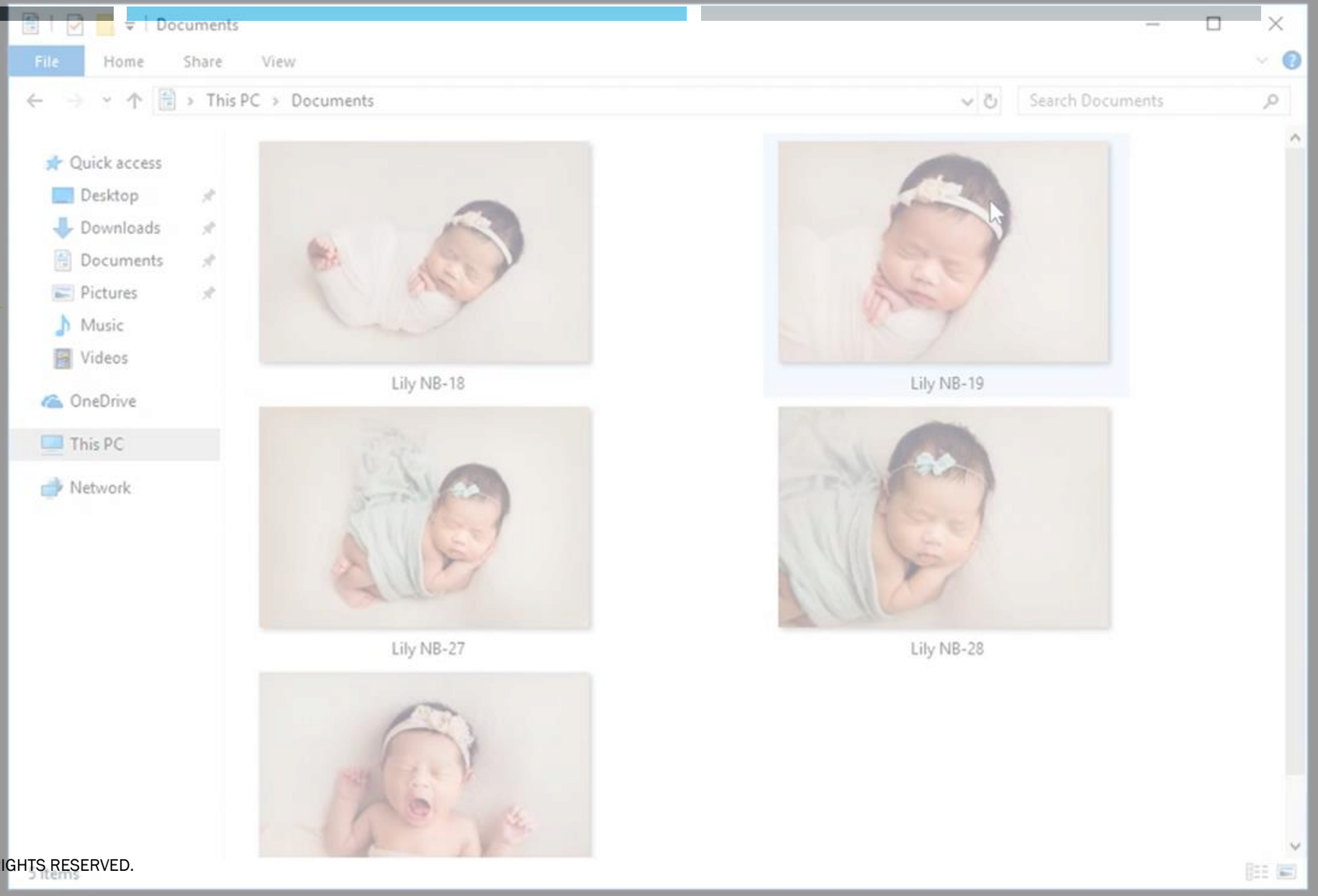
Detecting Lateral Movement



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DEMO

https://www.youtube.com/watch?v=OwJSZT2U4kM



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Conclusion

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Summary

- Endpoint Security
  - What & Why?
- Threat Actors
- Markets & Challenges
- Examined how AV works
- Understand the Ransomware Threat
- Explored Anti-ransomware



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Conclusions & Takeaways

- Importance of multi-faceted defence of the Endpoint.
- Extensive Backend Infrastructure
- Threat Landscape is constantly evolving.
- Endpoint Security more important today than ever.
- No one vendor has all the solutions.



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Thank you

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Engineering

Building Solutions for GA

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Timeline



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ENGINEERING PRACTICES



Agile/Scrum

Daily Standup

2-week Sprint

6-week PSI (Potentially Shippable Increment)

End-of-Sprint Demo

Sprint Planning

Proxy Product Owner



Test Driven Development

Pair Engineers with SDET

Automated Functional Tests

Integrated into build

1000+ Test Cases



Continuous Build

Daily/Weekly/On-demand (PR)

Integrated Tests



Virus Labs

Test against specific Malware

Remediate Top-10 Ransomware

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Performance Engineering

Ensuring ‘cost’ is within bounds

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Performance Engineering



Endpoint is ‘highly’ sensitive to resource utilization

Active monitoring costs

e.g. File copies



EPP should be in the background

Operation imperceptible to the user



Public Tests rule the day

AV-TEST

AV-Comparisons

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Performance Engineering



Real-world Scenarios

Install Browser

Install LibreOffice

Uncompress operation



Quantitative Loads

Determine if performance is linear or potentially exponential

Variables = No. of Files, Size/Type of Files, Directory depth/breath

Scientific Approach – i.e. modify 1 variable at a time.



User Profile

Define typical profiles for light, medium & heavy users

e.g. Browsing & Streaming, MS Office user, Software Engineer



Windows ADK

Internal Standard

Common Use-cases

(e.g. browser load, application launch, boot-up, restart, etc.)

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Performance Engineering



Manage Performance Metrics

Capture & Store performance data

Graphically display results



Performance Trends

Track performance data over time.

Determine general improvements

Capture anomalies/outliers



CI/CT

Continuous Testing

Integrated into build

Hourly/daily/weekly builds

Show trend

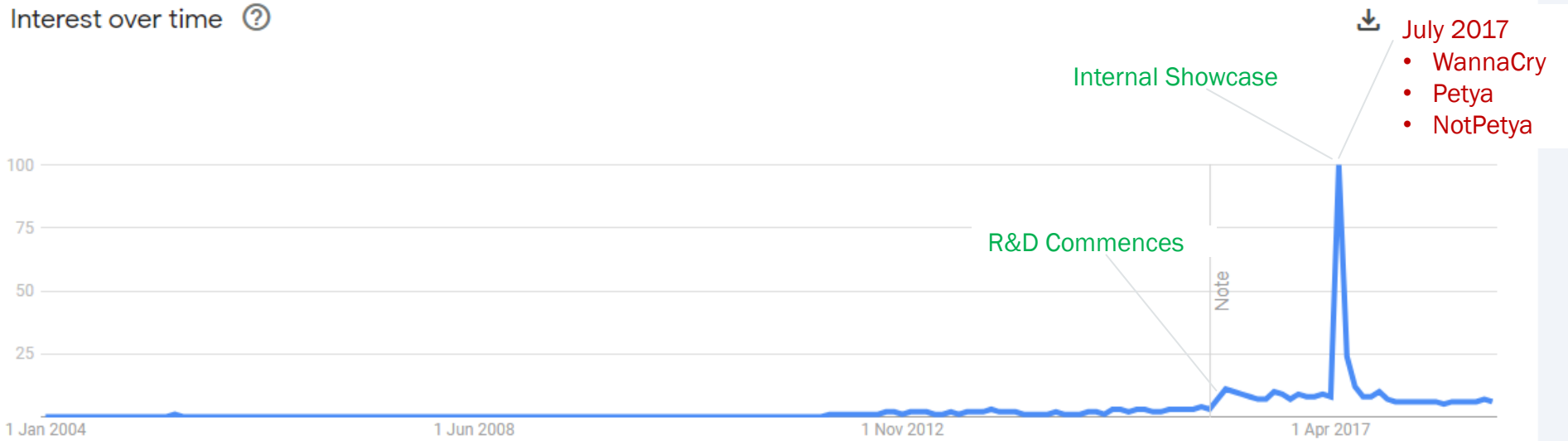
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ransomware
Search term

+ Compare

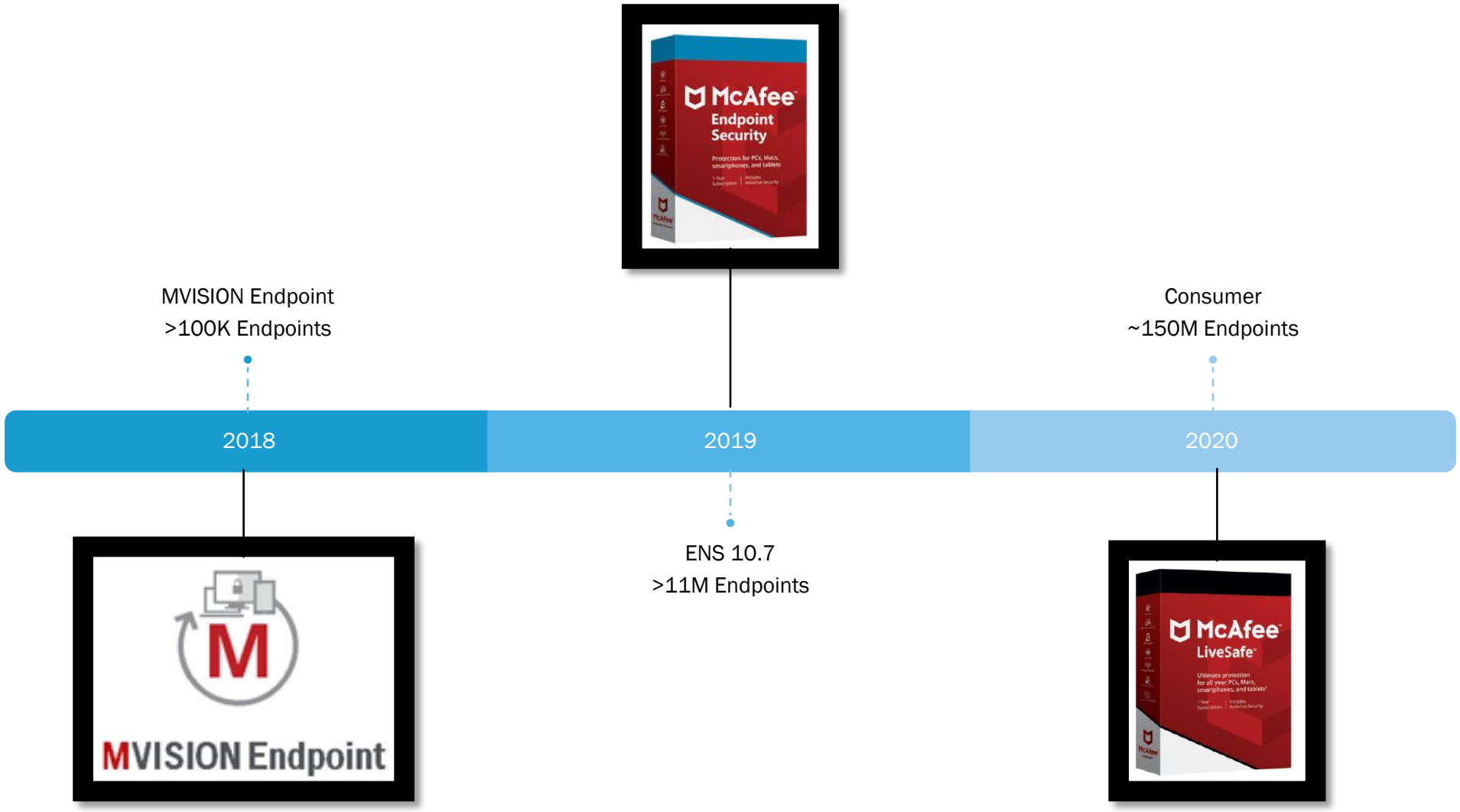
Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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Integration Timeline



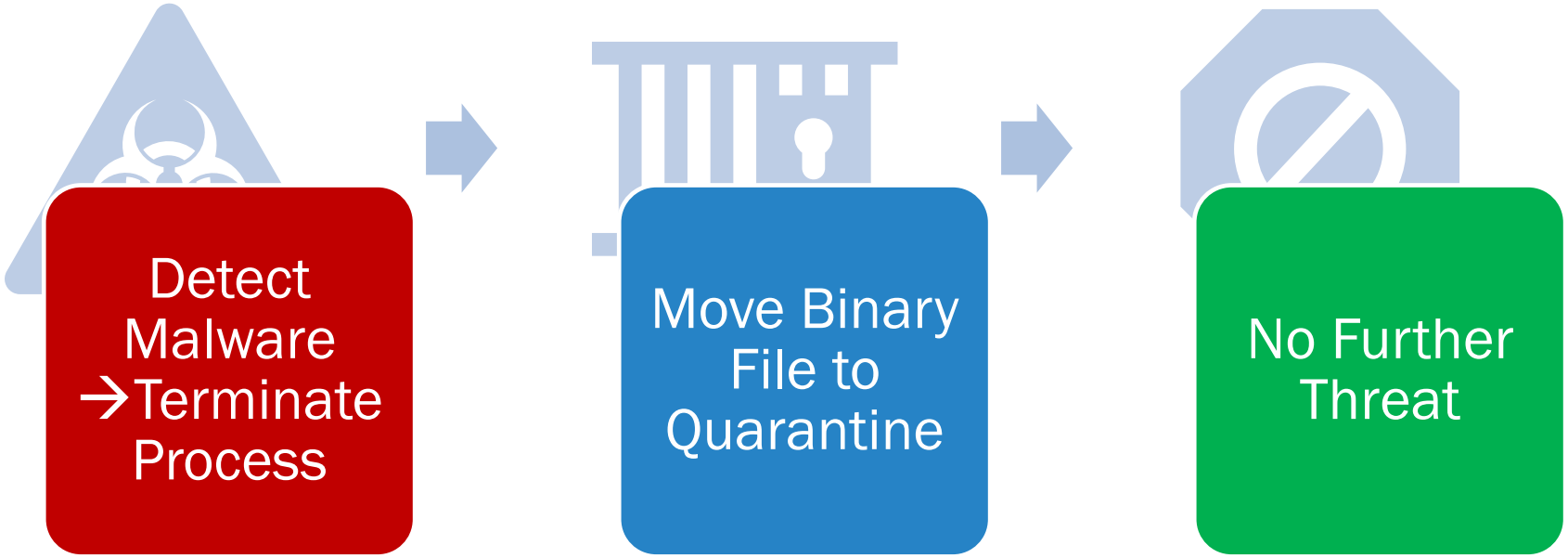
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Fileless Remediation

Breaking the Execution Chain

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Normal AV



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Systems

System Tree

Threat Event Log: Details — Previous | Next

Threat Handled:	True
Analyzer Detection Method:	MVISION Endpoint

Events received from managed systems

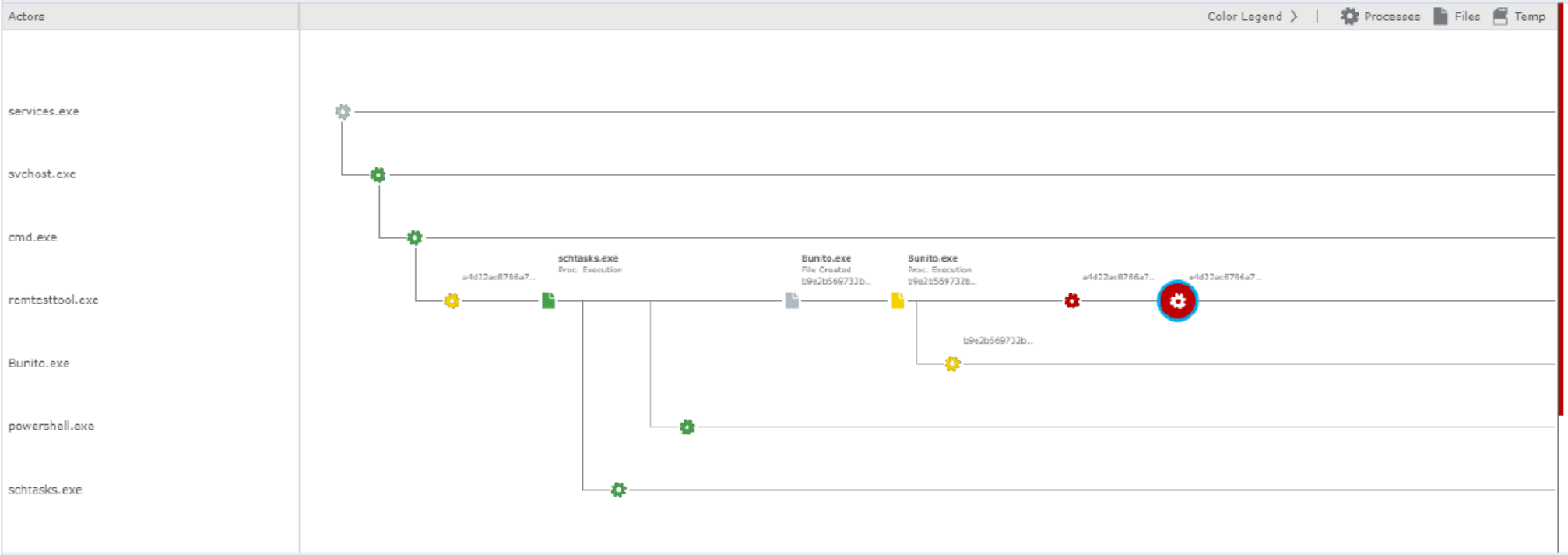
Event Description:	Infected file was moved to quarantine area.

Endpoint Security

Adaptive Threat Protection Events

Web Control Events

Story Graph (Trace Summary)



Event Details

Process terminated

Target Name	C:\Users\Ted\Desktop\deep_remediation_sample\remtesttool.exe
Reputation	Known Malicious
Reputation Score	1
PID	2964:132158147741768607

Action Taken
Threat has been quarantined

SHA-256
a4d22ac8786a75350f8ce8cc9f0f493fae227a6ff1e68945b23735f413785f49

MD5
4bf25e06e9e6c86e797da9a0cbc72fbb

Command Line Parameter
.\remtesttool.exe -sb -ib -wb -r -w1 18000 -db -rb -w2 180000

MVISION Endpoint

Target MD5 Hash:	4bf25e06e9e6c86e797da9a0cbc72fbb
Target SHA-256 Hash:	a4d22ac8786a75350f8ce8cc9f0f493fae227a6ff1e68945b23735f413785f49

Related Items

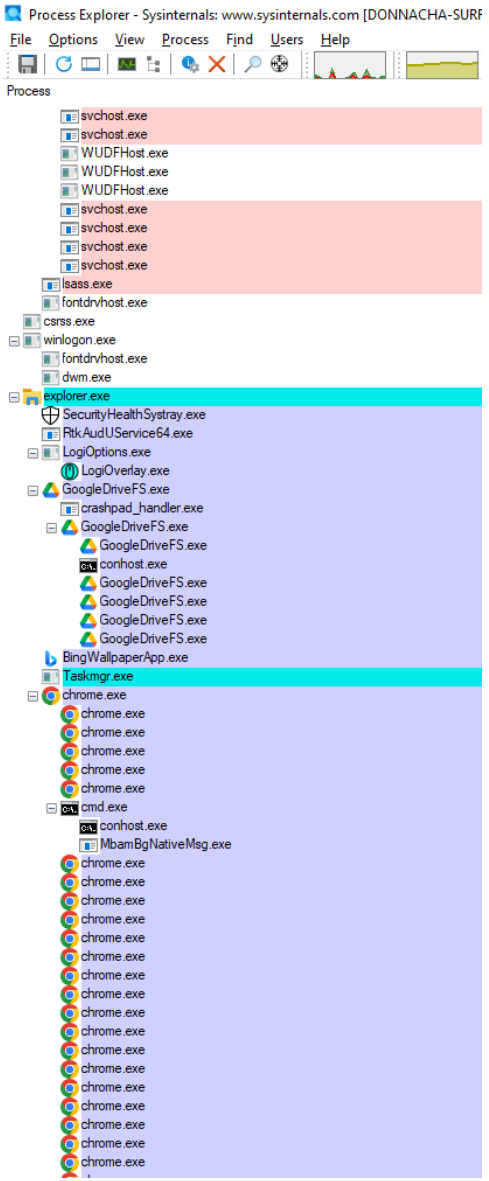
Go to related Systems

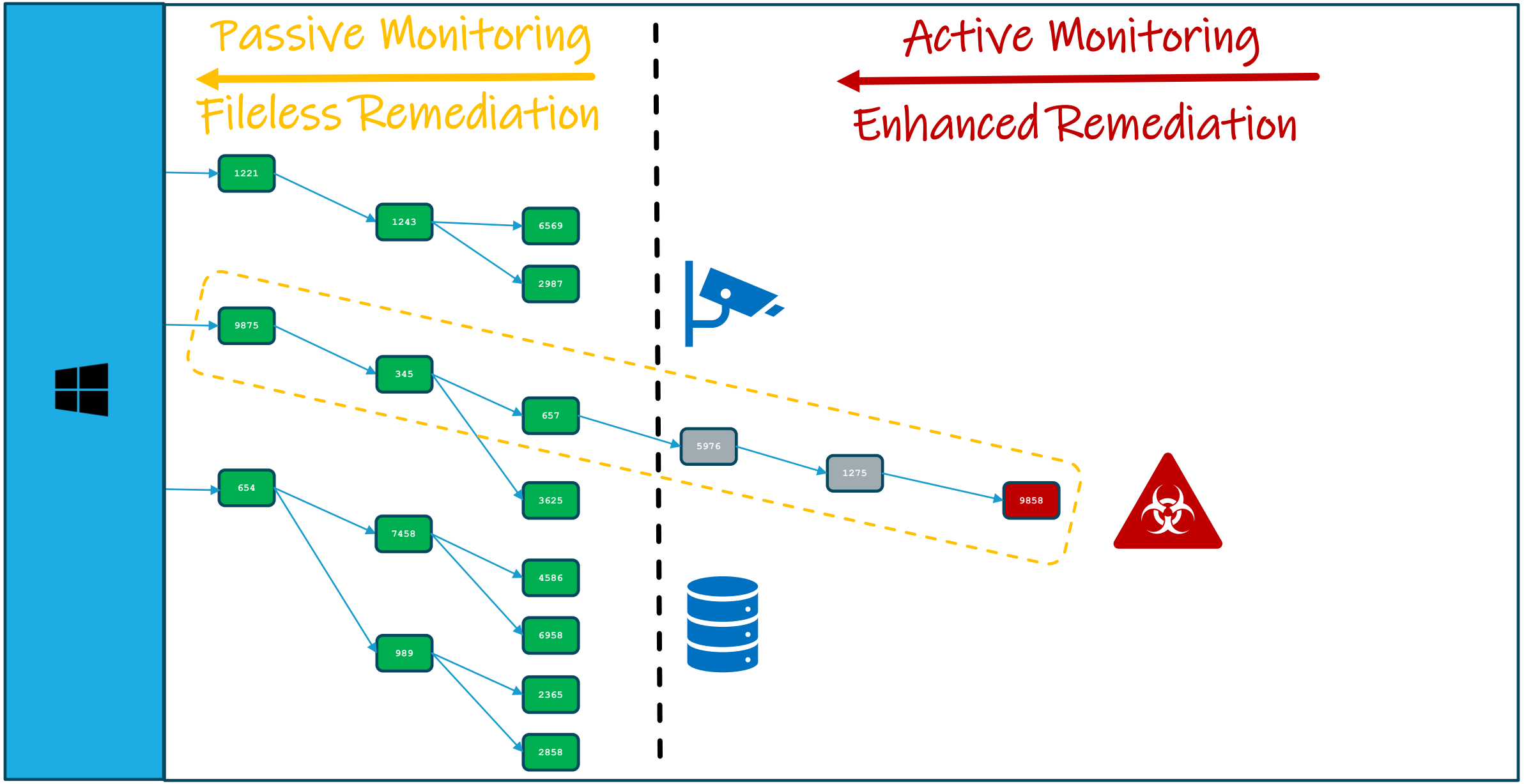
Actions | Show Source Systems | Show Target Systems

Back | Close

Process Tree Hierarchy

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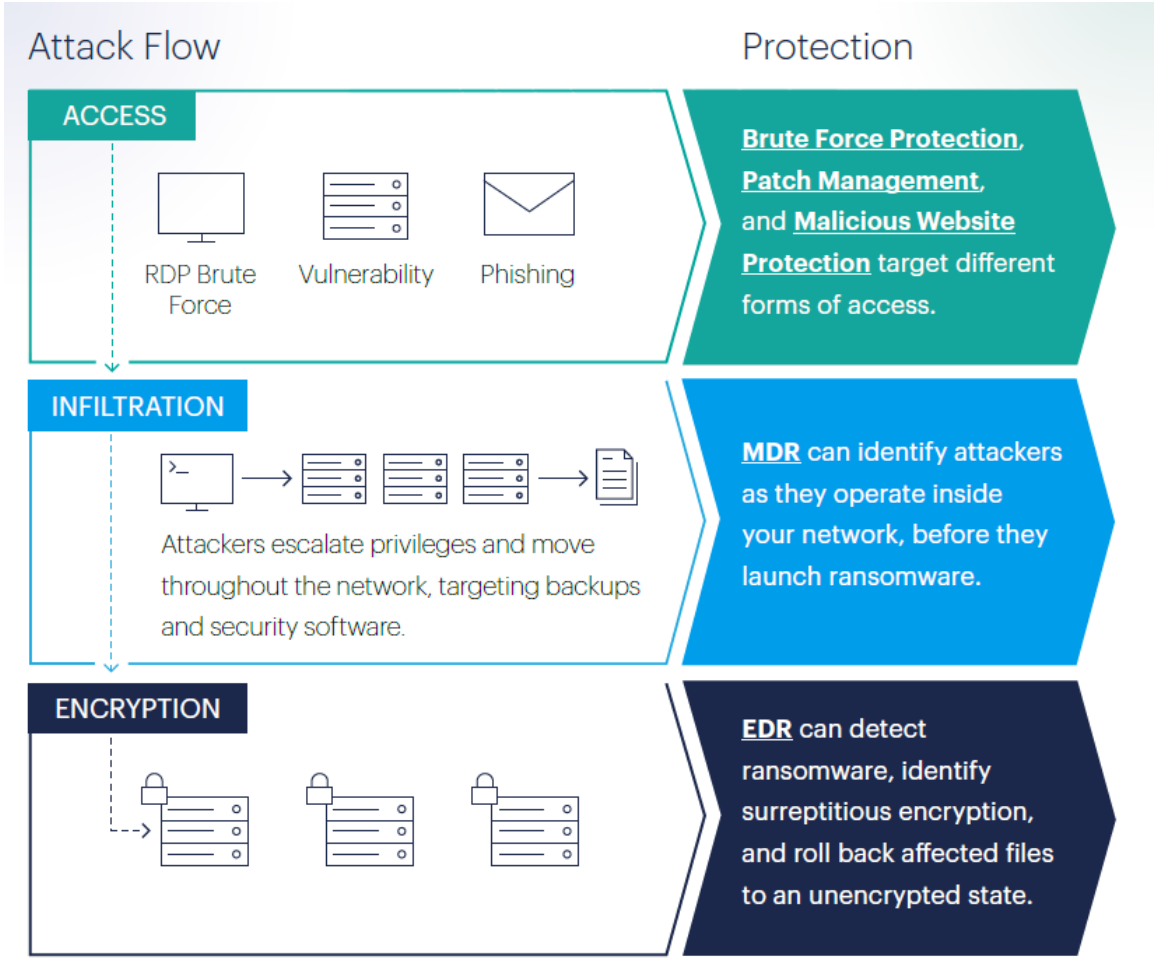




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How Ransomware Works

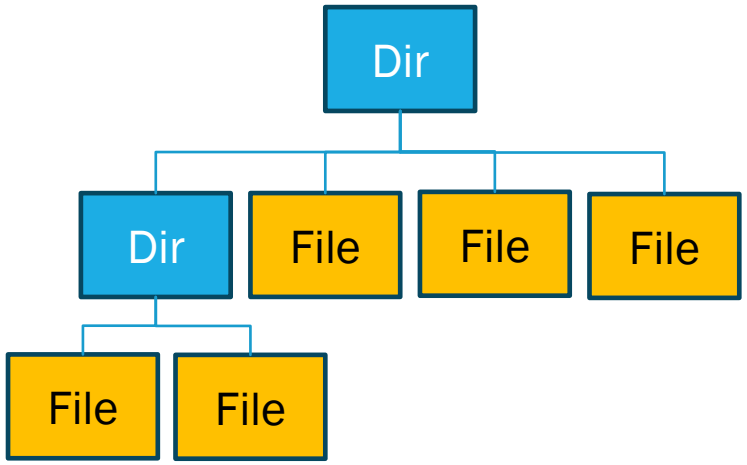


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Image Credit: *Malwarebytes – State of Ransomware Report 2023*

How Ransomware Works

Iterate over the File System



*Focus on typical document locations
*Target office docs

Pseudo Code

Open File System

- Open Directory
- Read File
- Write Encrypted File
- Delete Original
- Loop

Loop

What's the Problem?

Zero-day!

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Future Innovations



Deep Monitor ‘Greyware’



Track low-level operation of Malware AND Goodware



Big Data → Machine Learning → AI

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MVISION ENDPOINT

- Backstory
  - Circa 2017-2018
  - Stalled ENS 10.7 Adoption
  - Defender Shipped with Windows 10
- Goals
  - Lightweight & Performant
  - New UX
  - Work alongside AND manage MS Defender
  - Single pane of glass
  - Regular, monthly updates



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